

Skip-gram

Mikolov et al. 2013 word2Vec

Learn 2 vectors per word

word vec

Context Vector

Try to predict context from word

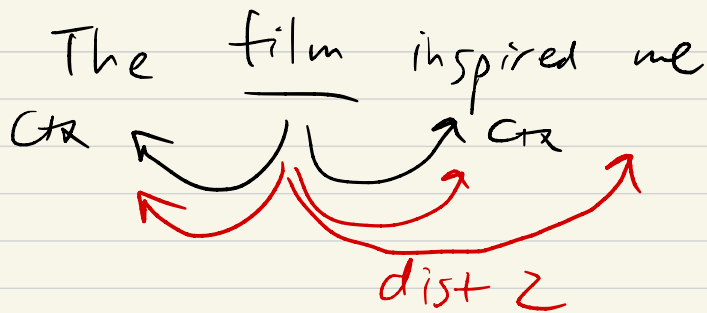
Input Corpus of text

Output $\bar{v}_w, \bar{c}_w$ for each word in vocab

Hyperparameters: d dimension (~ 300)

Window size K

$K=1$ word $K=2$



(word = film, context = The)
 $x =$ $y =$

(film, inspired)

(film, me)

Corpus, $K \Rightarrow$ set of pairs

$K=1$ 100,000 words

$\Rightarrow 200K$

Model skip-gram

$P(\text{context} = y \mid \text{word} = x)$

$$= \frac{e^{\vec{v}_x \cdot \vec{c}_y}}{\sum_{y \in V} e^{\vec{v}_x \cdot \vec{c}_y}}$$

dist over contexts

Parameters: $\mathcal{V}$ word vectors
 $|\mathcal{V}| \times d$
10,000 300

Context vectors $\mathcal{C}$
 $|\mathcal{V}| \times d$

randomly initialize

Train:
 $(\overset{x}{\text{word}}, \overset{y}{\text{context}})$

minimize $-\log \sum_{(x,y)} P(\text{ctx} = y | \text{word} = x)$

Ex Corpus = I saw $k=1$
 vocab = {I, saw} $d=2$

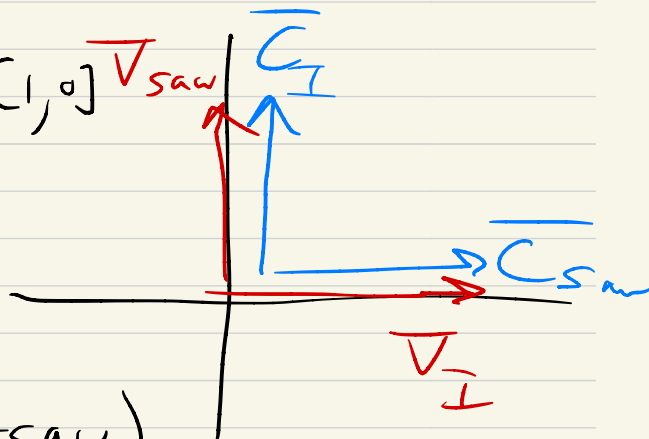
Assume: $\bar{V}_I = [1, 0]$ $\bar{V}_{saw} = [0, 1]$

① Let $\bar{C}_{saw} = [1, 0]$ $\bar{V}_{saw}$

$\bar{C}_I = [0, 1]$

what is

$P(\text{context} \mid w = \text{saw})$



2 outcomes: (I, saw)

3/4

$P(c = I \mid w = \text{saw}) =$

1/4

$$e^{\bar{V}_{saw} \cdot \bar{C}_I} = e$$

$$e$$

$$\frac{e^{\bar{V}_{saw} \cdot \bar{C}_I} + e^{\bar{V}_{saw} \cdot \bar{C}_{saw}}}{e^{\bar{V}_{saw} \cdot \bar{C}_I} + e^{\bar{V}_{saw} \cdot \bar{C}_{saw}}} = \frac{e + e}{e + e} = 1$$

$$P(c=\text{sam} | I=\text{sam}) \approx \frac{1}{4}$$

④ How can we change $\bar{c}$ to minimize loss?

$$\bar{c}_I = [0, 10]$$

$$= \frac{e^{10}}{e^{16} + 1} \approx 1$$

Problems

(N)

Suppose we have 100M corpus
vocab = 30k(V) d = 300

200M pairs - $O(Vd)$

(film, inspired)
Fix: (film, fish)

① Skip-gram with negative sampling (SGNS)

Take ^{each} (word, context) pair
as "real"

(word, sampled context) as
"fake"

Learn a classifier

$$P(\text{real} | y, x) = \frac{e^{\bar{v}_x \cdot \bar{c}_y}}{1 + e^{\bar{v}_x \cdot \bar{c}_y}}$$

SG = denominator of V

SGNS: denom $O(1)$